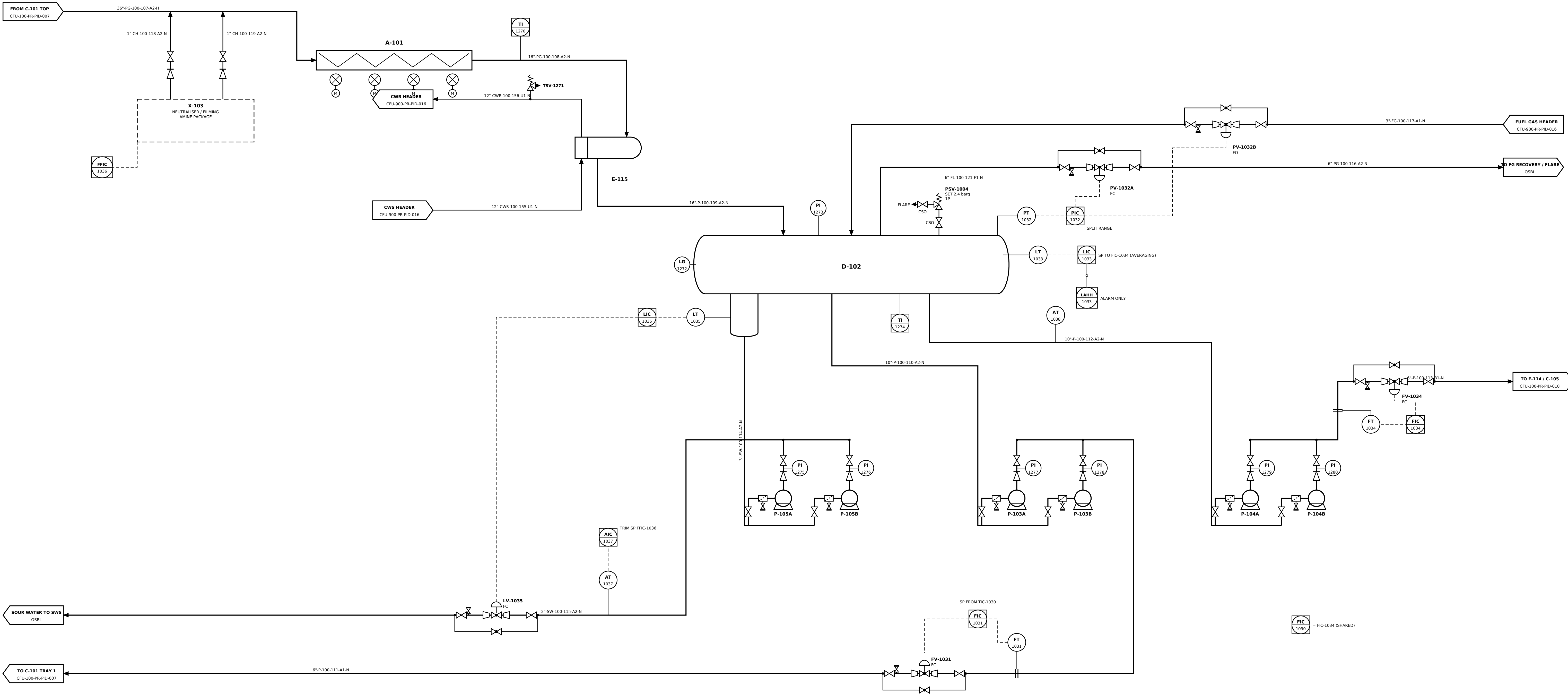


|                                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                              |                                                                                                                                            |                                                                                                                                                       |                                                                                                                                                   |                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>A-101</b><br>Atm. overhead condenser<br>1667 m2 bare; 3 bay(s) 6 m x 12 m, 6 fans<br>DESIGN 9.7 barg / 165 C<br>DUTY 33.75 MW; 6 x 55 kW fans<br>MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, TI optional) | <b>E-115</b><br>Atm. overhead trim condenser<br>740 m2; 2 shell(s)<br>DESIGN 3.5 / 7 barg / 90 C; CA 3 mm<br>DUTY 5.96 MW<br>SHELL: OH vapour/condensate / TUBE: Cooling water<br>MAT'L CS shell / TI tubes | <b>D-102</b><br>Atm. overhead reflux drum (3-phase)<br>ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m<br>DESIGN 3.5 barg / 160 C; CA 6 mm<br>MAT'L CS (HIC-resistant), 6 mm CA | <b>P-103A/B</b><br>Atm. reflux<br>RATED 159 m3/h @ 102 m, dP 7 bar<br>MOTOR: 55 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-104A/B</b><br>Unstabilised naphtha<br>RATED 180 m3/h @ 232 m, dP 16 bar<br>MOTOR: 160 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-105A/B</b><br>Overhead sour water<br>RATED 14 m3/h @ 51 m, dP 2 bar<br>MOTOR: 5.5 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>X-103</b><br>Neutraliser / filming amine package (atm OH)<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR: 0.75 kW<br>MAT'L SS316 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Neutraliser and filming amine injected upstream of A-101; pH control AIC-1037 trims FFIC-1036.
  - PIC-1032 split range: 0-50 % PV-1032B (FG make-up) closing, 50-100 % PV-1032A (off-gas) opening.
  - D-102 high-high level is a DCS alarm only (LAHH-1033); TSV on E-115 CW side, set per TSV-tyt.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |          |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|----------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |          |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |          |
| PIPING & INSTRUMENT DIAGRAM<br>C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102    |            |                          |            |          |
| DISCIPLINE                                                                  |            | STAGE                    | SCALE      | SHEET    |
| PROCESS                                                                     |            | FEED                     | NTS        | 10 OF 17 |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |          |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |          |
| DRAWING NUMBER                                                              |            |                          |            | REV      |
| CFU-100-PR-PID-009                                                          |            |                          |            | A        |